1 Question (15 points)

What are the outputs of the following programs?

a.)
```
public static void main(String[] args) {
   for (int i = 1; i <= 5; i++)
      printStar(i);
   for (int i = 5; i >= 1; i--)
      printStar(i);
}
public static void printStar(int n) {
   for (int i = 1; i <= n; i++)
      System.out.print("*");
   System.out.println();
}
```

b.)
```
int x = 4;
while (x < 10){
   x++;
   if (x > 8){
      break;
   }
   System.out.print(x);
}
```

c.)
```
int sum = 0;
for (int i = 5; i >= 1; i--){
   for (int j = 5; j >= i; j--){
      sum += 1;
   }
   System.out.println(sum);
}
```

2 Question (20 points)

Write a method **sum** to calculate the sum of a set of integers with their frequencies, which are stored in a file. The name of the file is passed as parameter and the method returns the sum of the numbers in the file.

```
public static int sum(String fileName)
```

The format of the file is as follows: The first line shows the number of integers stored in the file. Each line after the first line stores two numbers, the frequency and the number itself.

```
fileName -> input1.txt
3
2 22
1 2
3 9
Returns 2 x 22 + 1 x 2 + 3 x 9 = 73

fileName -> input2.txt
5
1 1
1 2
1 3
1 4
1 5
Returns 1 x 1 + 1 x 2 + 1 x 3 + 1 x 4 + 1 x 5 = 15
```

3 Question (15 points)

Write a program that rolls a 6-sided die until it gets two consecutive rolls in a row. As the program rolls the die it should print each value rolled, and then it should print a message at the end to indicate how many rolls were made. In the first example, notice that the program stops after rolling 3, 3.

```
5246133
Finished after 7 rolls!
```

```
1363524344
Finished after 10 rolls!
```

4 Question (20 points)

Write a method named **yardSale** that takes an initial amount of money and the number of purchase trials as its parameter, and returns the amount of money the user ends up with after a series of purchases.

Your method should ask the user for a series of potential purchases specified by a unit price per single item and the quantity of that item. The user buys everything that does not exceed his current amount of money.

A purchase is considered a **good deal** if the unit price is under 10 TL, **a message should be printed** when such a purchase is made. In addition, your method should print the **total quantity of items purchased** and the total cost of the **most expensive** item purchased. With an initial money of 50.75 TL and 4 purchase trials:

Price? <u>8.75</u>
Quantity? <u>2</u>
What a deal! I'll buy it.
Remaining money: 33.25

Price? <u>11.50</u>
Quantity? <u>3</u>
Remaining money: 33.25

Price? <u>5.10</u>
Quantity? <u>5</u>
What a deal! I'll buy it.
Remaining money: 7.75

Price? <u>0.75</u>
Quantity? <u>8</u>
What a deal! I'll buy it.
Remaining money: 1.75

Total quantities purchased: 15
Most expensive purchase: 25.5

5 Question (15 points)

In this question you will read an integer n from the user and then read n more integers. You should find the smallest (min) and the largest number (max) entered by the user and calculate $\min^{\max}$ without using pow method from the Math library. You are given the main method. **You will write a method named calculatePower which takes a Scanner parameter.**

1. First, you will ask the user how many numbers she wants to enter. You should do this inside the method.
2. Then user will enter the numbers as shown in the example. You may assume that the user will enter integers between 0 and 100.
3. calculatePower method will **print the minimum and the maximum of the grades** entered by the user.
4. calculatePower method will calculate and **return the minimum number raised to the power maximum**.

How many numbers:
<u>4</u>
<u>4 2 6 3</u>
Minimum: 2
Maximum: 6
Answer: 64.0

6 Question (15 points)

Write a method named vowelsMinusConsonants which takes a string parameter and returns the difference between the number of vowels and the number of consonants in the string s.

A call to vowelsMinusConsonants by parameter “acababe” should return 1. There are 4 vowels and 3 consonants in a.